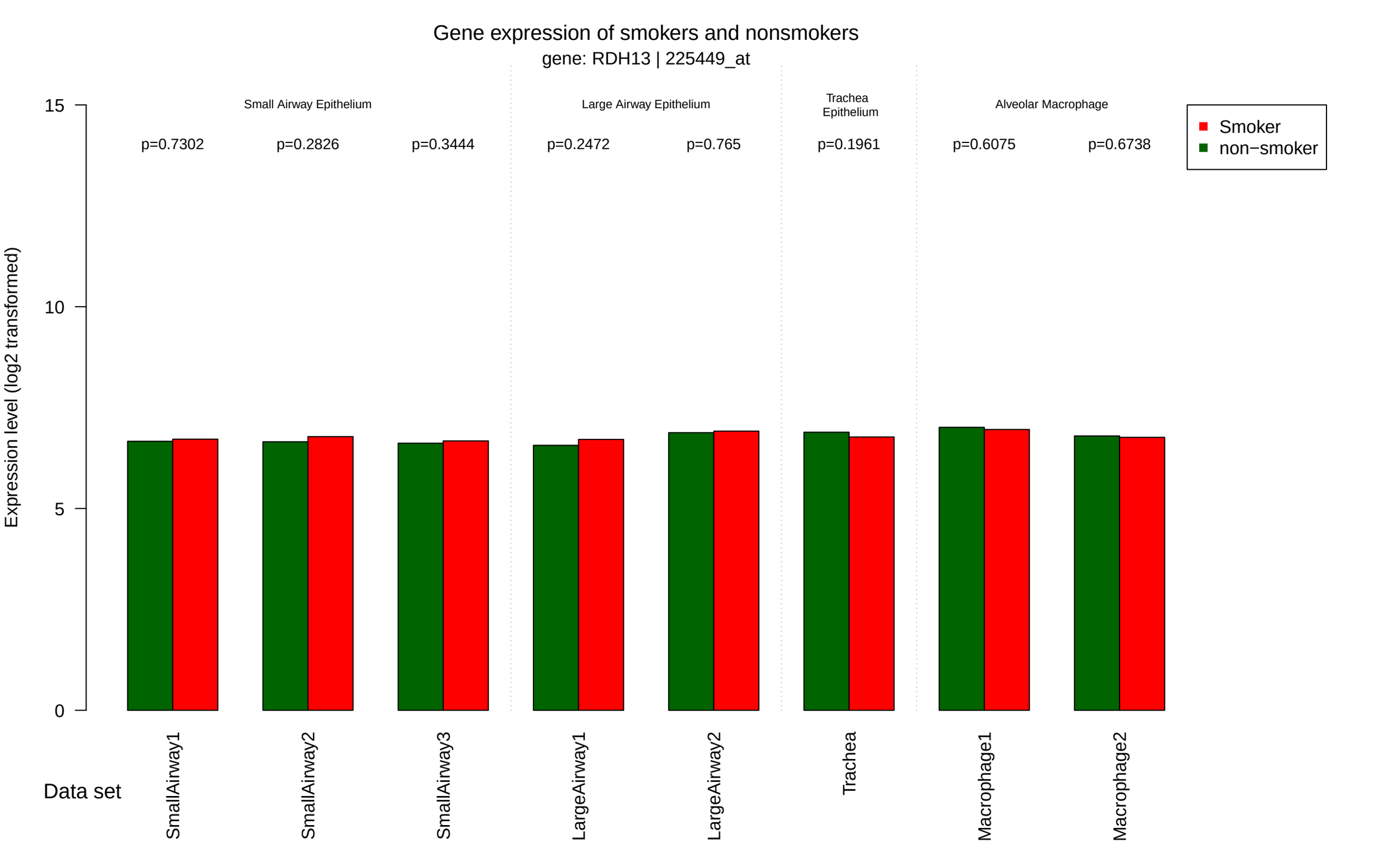
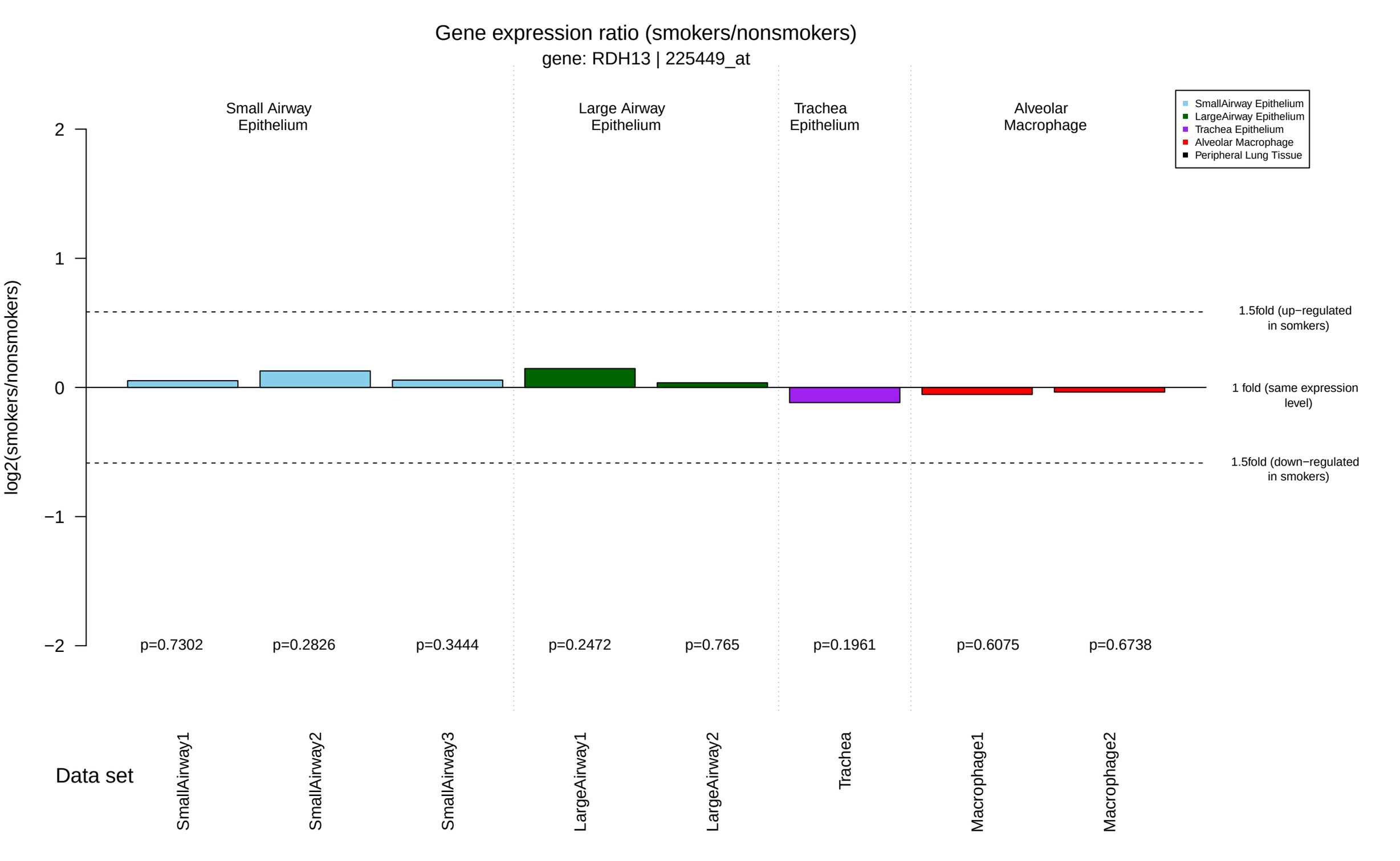
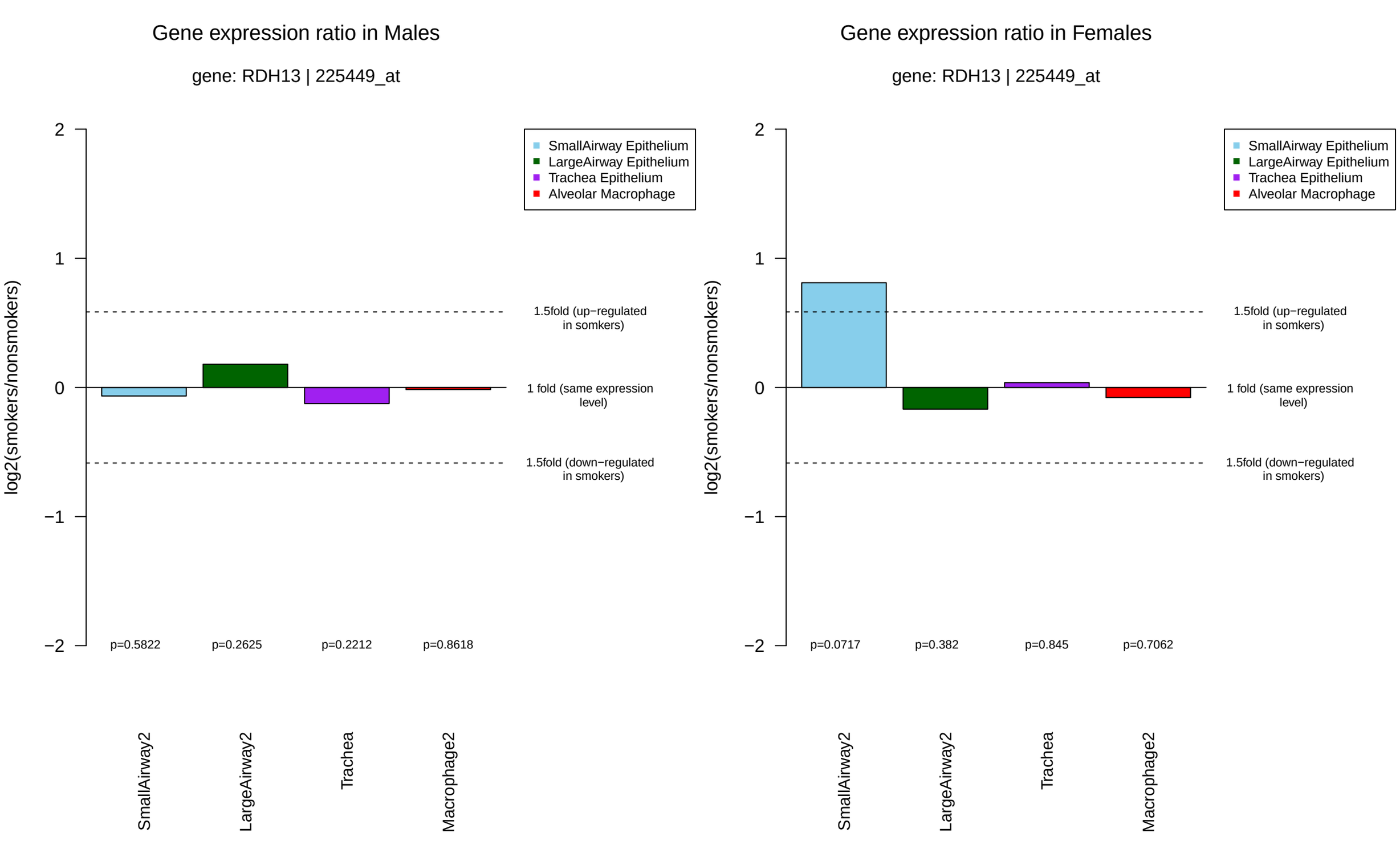
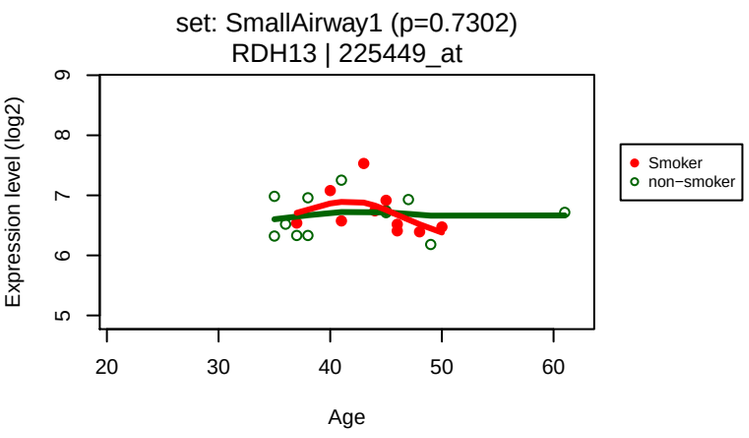
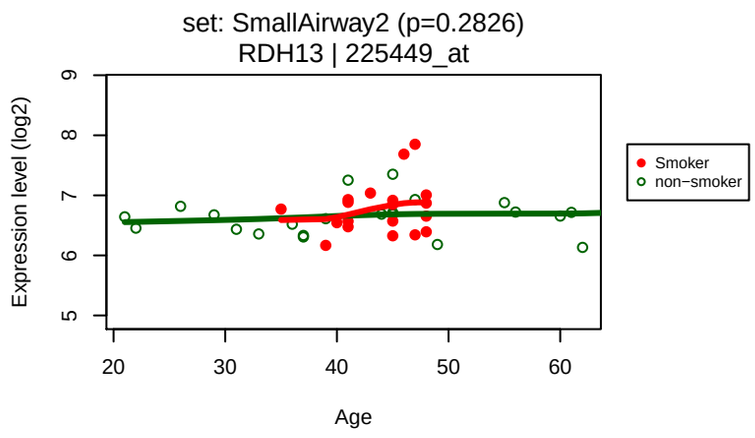
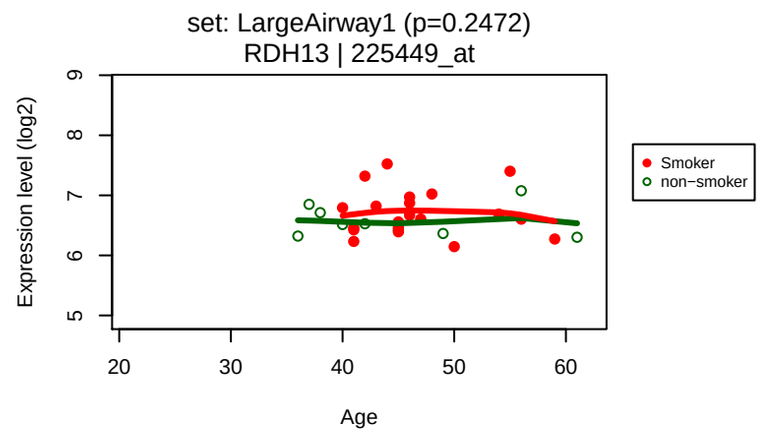

Gene expression of smokers and nonsmokers
gene: RDH13 | 225449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7302
p=0.2826
p=0.3444
p=0.2472
p=0.765
p=0.1961
p=0.6075
p=0.6738
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

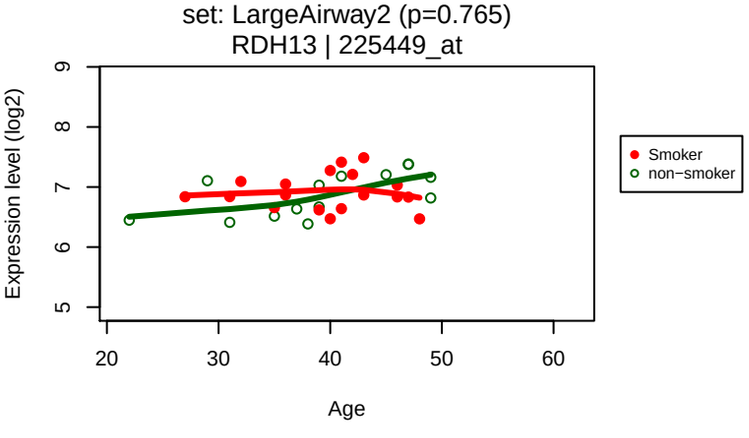

Gene expression ratio (smokers/nonsmokers)
gene: RDH13 | 225449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7302
p=0.2826
p=0.3444
p=0.2472
p=0.765
p=0.1961
p=0.6075
p=0.6738
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

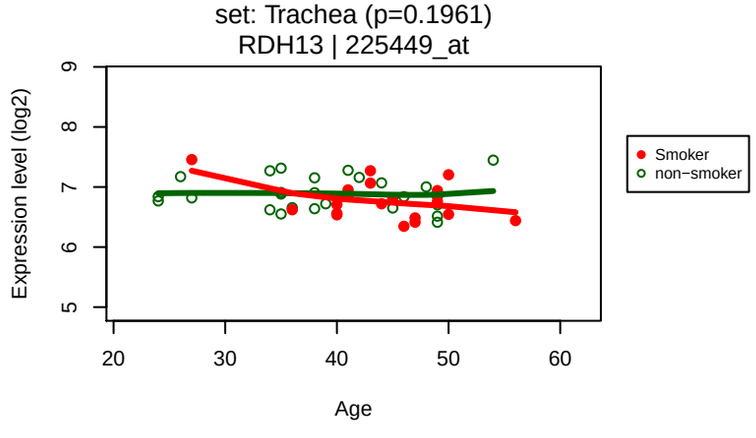

Gene expression ratio in Males
gene: RDH13 | 225449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5822
p=0.2625
p=0.2212
p=0.8618
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH13 | 225449_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0717
p=0.382
p=0.845
p=0.7062
SmallAirway2
LargeAirway2
Trachea
Macrophage2

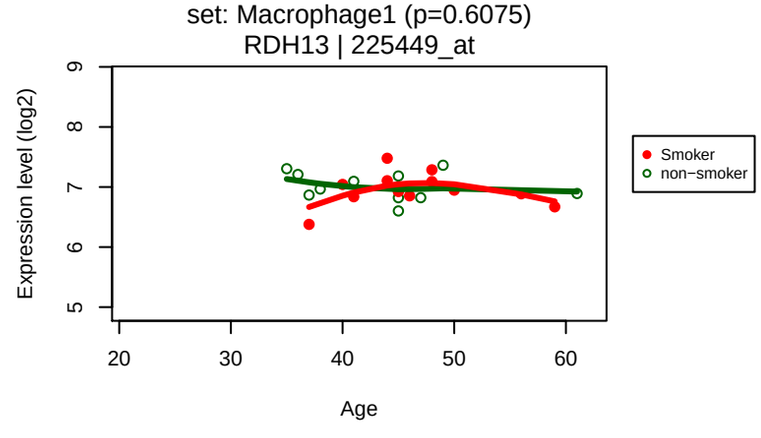

set: SmallAirway1 (p=0.7302)
RDH13 | 225449_at
Expression level (log2)
Age
Smoker
non−smoker

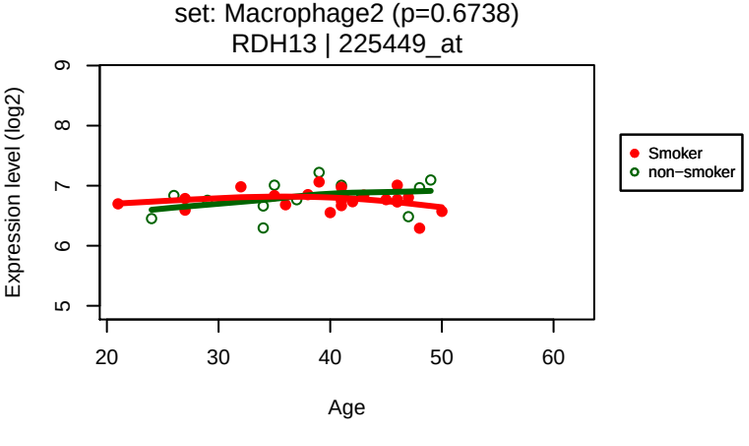

set: SmallAirway2 (p=0.2826)
RDH13 | 225449_at
Expression level (log2)
Age
Smoker
non−smoker

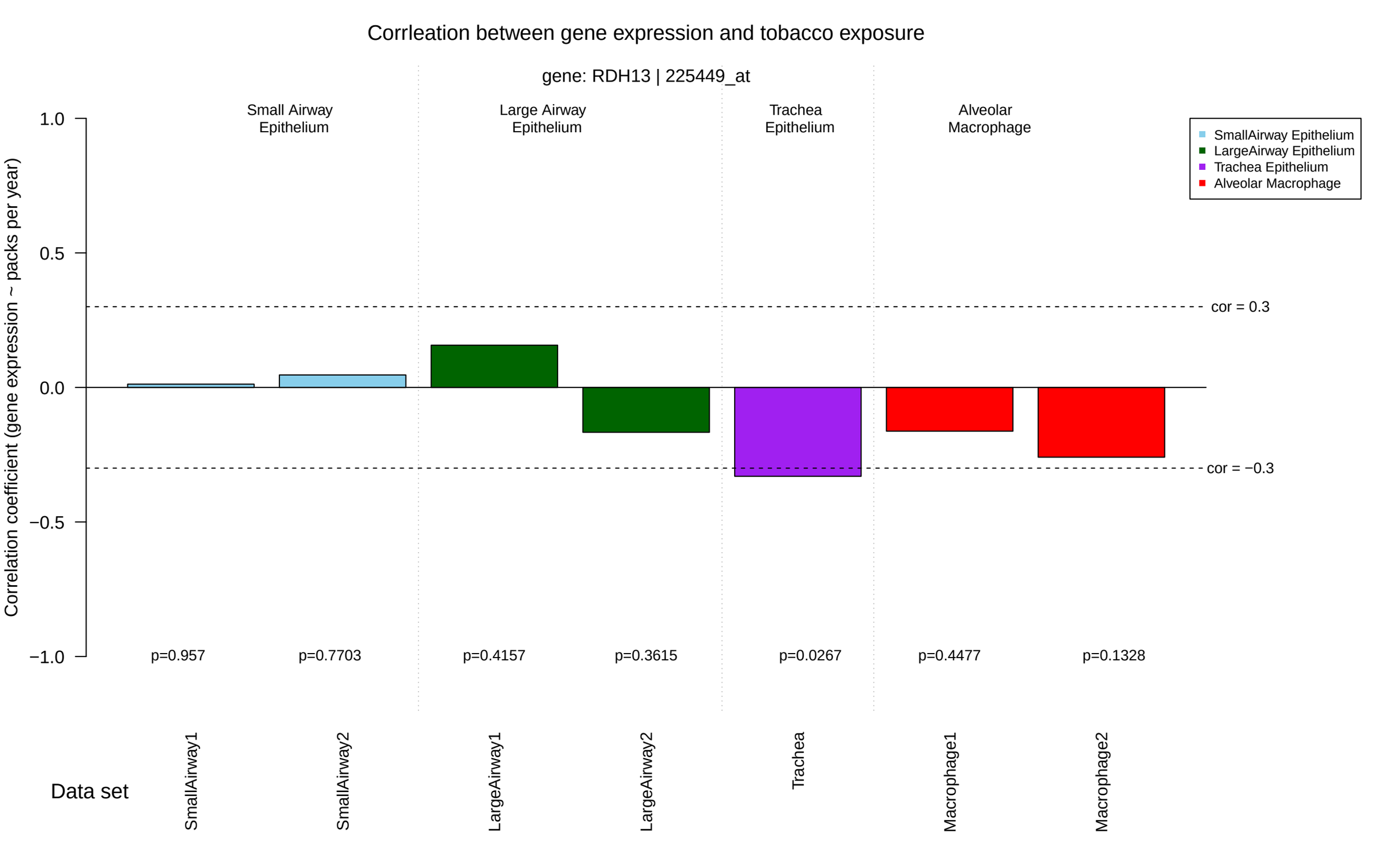

set: LargeAirway1 (p=0.2472)
RDH13 | 225449_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.765)
RDH13 | 225449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1961)
RDH13 | 225449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6075)
RDH13 | 225449_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6738)
RDH13 | 225449_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH13 | 225449_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.957
p=0.7703
p=0.4157
p=0.3615
p=0.0267
p=0.4477
p=0.1328
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2